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Editors
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Dear Editors:

Please consider the enclosed Letter for publication in *Nano Letters*: "Dynamically Heterogeneous Mesoscale Arrest in a Protein Polymer with Reversible $\beta$-Sheet-like Order."

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**Significance.** Whether a soft material has arrested is usually read off its structure: a scattering profile that has grown enough is taken to be a gel. In the sequence-defined elastin-like polypeptide $(VPAVG)_{30}$ the structure does not decide it. Small-angle X-ray photon correlation spectroscopy (SA-XPCS) separates the assembly that SAXS records from the arrest that follows. Four loadings were ramped at 0.1 °C/min to a final temperature between 28.00 and 28.75 °C and held there. All four assemble, their low-$Q$ intensity growing throughout the hold, but only those above 28.00 °C arrest within it, and SAXS cannot tell the states apart: at the onset of arrest at 28.25 and 28.75 °C the profile lies within 18% at every $Q$ of the 28.00 °C profile, which is still mobile after 6500 s. A quarter of a degree decides it, not an amount of structure. Arrest also reaches the largest length scales first: the fast-decay fraction $f$ rises monotonically with $Q$ in every group of every arresting loading, and at 28.50 °C it falls from 0.80 to 0.07 at 175 nm within an hour. Temperature-dependent SAXS/WAXS on the same construct resolves a reversible transition to a hierarchical assembly carrying $\beta$-sheet-like peaks at 1.38 and 0.72 $Å^{-1}$ that vanish on cooling, with seven consecutive 6–34 °C cycles recovering the low-temperature profile to within 9%.

**Novelty.** Prior work on gel-forming ELPs established arrested phase separation and network morphology from time-averaged scattering and rheology. The dynamical signature of a fast mode whose time scale persists while its amplitude declines has also been seen in a thermoreversible nanoparticle gel [D. Bahadur *et al.*, *J. Chem. Phys.* **151**, 104902 (2019)], and the manuscript cites that agreement rather than claiming the phenomenology as new. Two things here are. First, a temperature series resolves assembly and arrest as separate steps and shows that temperature, not structure, sets the second; resolving the fast fraction in each $Q$ bin then shows arrest is ordered by length scale and follows the transfer of material out of the mobile population to an arrested fraction above 0.9. Second, one rescaling of the waiting time, $\exp(\Delta T/T_s)$, collapses all four holds onto a single master curve with $T_s = 0.22 \pm 0.02$ K, and the same $T_s$ to within 13% in every one of the five $Q$ bins. That is a stronger statement than a rate: the sample passes through the same sequence of dynamical states at every temperature and every length scale measured, and temperature sets only the pace. A scaling temperature of a few tenths of a kelvin is also too steep for a single contact, which places ELP arrest in the same cooperative regime as thermoreversible colloidal gels, where the identical analysis gives 0.33 and 0.45 K. The decomposition needs microsecond framing: the fast relaxation is millisecond-scale, so a kilohertz acquisition returns only the slow component. We are equally explicit about the limits. The structural and dynamical observations come from separate experiments, so no causal link is claimed. The $\beta$-sheet assignment rests on WAXS peak positions rather than spectroscopy. The 1.6 s maximum delay bounds a fraction arrested on that scale, not an infinite-time nonergodicity factor. The scaling rests on three arresting temperatures spanning half a degree, and the collapse is approximate rather than exact.

**Impact and urgency.** The work spans sequence-defined nanomaterials, protein self-assembly, nonequilibrium soft matter, and coherent X-ray characterization, and gives a nanoscale structure–dynamics framework for responsive protein materials whose $\beta$-sheet-like order switches thermally without permanent aggregation. Because the behavior

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is sequence-encoded, the strategy transfers to other programmable protein polymers. The result rests on a capability that is not widely available. The XSPA-500k at 8-ID-I frames at 50 kHz, among the fastest in routine operation at a storage ring. The millisecond-scale fast relaxation reported here would fall at or below the shortest delay of a conventional kilohertz acquisition, and neither the length-scale decomposition nor the scaling collapse would be visible at all. High-speed detectors and upgraded storage rings have only recently made this practical on weakly scattering biological samples, so a concise, carefully bounded demonstration is timely. The reduced data and all analysis and figure-generation code are already public on GitHub (qzhang234/PubCode-2021-IDPP).

The manuscript is original, is not under consideration elsewhere, and has been approved by all authors. The authors declare no competing financial interest; AI-tool use is disclosed in the Acknowledgments. Thank you for your consideration.

Sincerely,

Qingteng Zhang, Ph.D., on behalf of all authors

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# Suggested Reviewers

The following researchers have relevant expertise in elastin-like polypeptides, $\beta$-sheet-forming protein and peptide assemblies, small-angle X-ray scattering, and X-ray photon correlation spectroscopy.

**Bradley D. Olsen**
Alexander and I. Michael Kasser (1960) Professor of Chemical Engineering, Massachusetts Institute of Technology
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Expertise: elastin-like polypeptide gels, associative protein polymers, arrested phase separation, and neutron spin-echo measurement of relaxation in protein hydrogels.

**Darrin J. Pochan**
Professor of Materials Science and Engineering, University of Delaware
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Expertise: peptide and protein self-assembly, $\beta$-sheet hydrogel networks, and small-angle scattering and rheology of soft biomaterials.

**Andrei Fluerasu**
Lead Beamline Scientist, Coherent Hard X-ray Scattering, National Synchrotron Light Source II, Brookhaven National Laboratory
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Expertise: X-ray photon correlation spectroscopy and the microscopic dynamics of gels and arrested soft matter.

**Theyencheri Narayanan**
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Expertise: small-angle X-ray scattering and X-ray photon correlation spectroscopy of colloids and soft matter.